

Supporting Information

Thermodynamic Modeling of Lithium Extraction from Synthetic Brine Using Deep Eutectic Solvents: A PC-SAFT Approach

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Group contribution method for calculation of Gibbs energy of formation

The details of the group contribution method for calculation of Gibbs energy of formation for organic components including TBAC, DA, DES, RLi and RNa is introduced. Table S1 shows the contribution of each group along with their reference number.

Table S1: The values of group contribution for calculation of Gibbs free energy of formation.

Group	Contribution	Reference
–CH ₃	-43.96	[1]
–CH ₂ –	8.42	[1]
–COOH	-387.87	[1]
–COO –	-301.95	[1]
> N –	163.16	[1]
–Cl	-64.31	[1]
Li	-250.254	[2]
Na	-199.801	[2]

Gibbs energy of formation for each compound was calculated by Equation S1 [1]. For Li and Na values in the table are used directly from reference [2].

$$\Delta G_{f,298}^0 = 53.88 + \sum n_i \times C_i \quad (S1)$$

where, C is the contribution of group i and n is the number of times group i appears in a component.

The groups appear in decanoic acid (CH₃(CH₂)₈COOH) are presented in Table S2.

Table S2: The groups appear in decanoic acid.

Group	n_i	C_i
–CH ₃	1	-43.96
–CH ₂ –	8	8.42
–COOH	1	-387.87

The Gibbs energy of formation for Decanoic acid:

$$\Delta G_{f, 298_{DA}}^0 = 53.88 - 43.96 + 8 \times 8.42 - 387.87 = -310.59 \frac{\text{kJ}}{\text{mol}}$$

The groups appear in Tetrabutylammonium chloride (TBAC) ($(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2)_4\text{NCl}$) are presented in Table S3.

Table S3: The groups appear in TBAC.

Group	n_i	C_i
$-\text{CH}_3$	4	-43.96
$-\text{CH}_2 -$	12	8.42
$-\text{N} <$	1	163.16
$-\text{Cl}$	1	-64.31

The Gibbs energy of formation for TBAC:

$$\Delta G_{f, 298_{TBAC}}^0 = 53.88 + 4 \times -43.96 + 12 \times 8.42 - 64.31 + 163.16 = 77.93 \frac{\text{kJ}}{\text{mol}}$$

The DES is composed of one TBAC and 2 Decanoic acid molecules, so:

$$\Delta G_{f, 298_{DES}}^0 = 77.93 + 2 \times -310.59 = -543.25 \frac{\text{kJ}}{\text{mol}}$$

The other two components in organic phase are RLi and RNa. These components are the same as DES but the hydrogen atom in one decanoic acid molecule is exchanged with Li or Na. So, there are two new components named dec-Li and dec-Na. Their groups are presented in Table S4.

Table S4: The groups appear in dec-Li and dec-Na.

Group	n_i	C_i
$-\text{CH}_3$	1	-43.96
$-\text{CH}_2 -$	8	8.42
$-\text{COO} -$	1	-301.95
Li	1	-250.254
Na	1	-199.801

The Gibbs energy of formation for dec-Li:

$$\Delta G_{f_{298_{dec-Li}}}^0 = (53.88 - 43.96 + 8 \times 8.42 - 301.95) - 250.254 = -474.92 \frac{\text{kJ}}{\text{mol}}$$

The Gibbs energy of formation for dec-Na:

$$\Delta G_{f_{298_{dec-Na}}}^0 = (53.88 - 43.96 + 8 \times 8.42 - 301.95) - 199.801 = -424.47 \frac{\text{kJ}}{\text{mol}}$$

The RLi is composed of one TBAC, one Decanoic acid and one dec-Li. Also, RNa is composed of one TBAC, one decanoic acid and one dec-Na. So, the Gibbs energy of formation for RLi and RNa can be calculated as follows:

$$\Delta G_{f_{298_{RLi}}}^0 = 77.93 - 310.59 - 474.92 = -707.58 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta G_{f_{298_{RNa}}}^0 = 77.93 - 310.59 - 424.47 = -657.13 \frac{\text{kJ}}{\text{mol}}$$

Experimental density data

The DESs were prepared in our previous work by heating method [3]. To measure the density of the DESs in this work, the precise quantities of DA and TBAC were determined using mass balance in accordance with TBAC|(1):DA(2) molar ratio. Specified masses of DA and TBAC were measured using a balance (Precisa XT220A) with $\pm 1 \times 10^{-4}$ g measurement uncertainty. Both components were mixed in a two-shell self-made flask, connected to a water circulator (Julabo F12) with ± 0.1 °C uncertainty to maintain temperature. Temperature was set at 60.0 °C and the solution was mixed using a magnetic stirrer at a speed of 1000 rpm. Mixing continued until a homogeneous and clear liquid was obtained. The mixture was left overnight to cool to room temperature, and, to check whether a solid phase appeared. Then, the density of the obtained DES and solution of 10%wt TOPO in DES was measured using an Anton Paar DMA HPM instrument according to ASTM D4052. The experimental data along with their uncertainties are tabulated in Tables S5 and S6, respectively. The uncertainty in measurement is reported in standard uncertainties u (0.68 level of confidence) and the source of uncertainty is from instrument resolution. All of the density data was measured in liquid phase and atmospheric pressure.

Table S5. Measured density of DES at various temperatures. The uncertainty in measurement for temperature is ± 0.01 K and for density is ± 0.1 kg/m³.

T (K)	Density (kg/m ³)
272.04	932.2
280.37	926.7
288.70	921.3
297.04	916.0
305.37	910.7
322.04	900.1
330.37	894.7

Table S6. Measured density of 10%wt TOPO in DES at various temperatures. The uncertainty in measurement for temperature is ± 0.01 K and for density is ± 0.1 kg/m³.

T (K)	Density(kg/m ³)
293.15	908.3
298.15	905.0
303.15	901.5
308.15	898.0
313.15	894.5
318.15	891.5
323.15	887.7
328.15	883.9
333.15	881.2
338.15	876.7
343.15	873.9

References.

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